

Template-based Extraction of Interbeat Intervals from In-Ear Heartbeat Sounds

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Abstract—Recent advancements in hearable technology have led to the development of multifunctional in-ear devices equipped with microphones that can improve audio experiences, assist with hearing and enhance speech communication. This paper introduces a template-based algorithm for extracting interbeat intervals (IBIs) from heartbeat sounds captured by acoustic in-ear microphones. The proposed method involves generating a template using the median waveform of the heartbeat sound, using this template for correlation-based peak detection, and detecting anomalies in the resulting peaks. The algorithm is evaluated on 39 participants in silence and noise, with ground truth provided by concurrent electrocardiography measurements. Results demonstrate that the template-based approach improves the accuracy of IBI extraction, achieving 4.68 milliseconds mean absolute error using a 300-second analysis window, which is an improvement of 55.13% from the benchmark method found in the literature. This study highlights the potential of template-based algorithms for heart rate variability analysis using acoustic in-ear microphones, broadening the applications of hearable technology for continuous non-invasive health monitoring.

Index Terms—digital health, human sensing, physiological signal processing, headphones, wearable technology

I. INTRODUCTION

The combination of wearable biosensing technology and advancements in machine learning has opened up avenues to monitor an individual's stress levels in real-time. By analyzing biosensor data, wearable devices can identify changes in autonomic nervous system activity associated with stress [1]. Various characteristics associated with psychological stress can be derived from heartbeat signals, as both the sympathetic “fight or flight” and parasympathetic “rest and digest” responses of the autonomic nervous system influence heart

activity across different time intervals. By precisely measuring the milliseconds between successive heartbeats, known as Interbeat Interval (IBI)s, numerous Heart Rate Variability (HRV) metrics can be calculated [2].

Wearable devices designed to be worn in or on the ear, also known as *hearables* or *earables*, have demonstrated efficacy in capturing physiological signals, including heartbeat sounds, directly from within the ear canal, making them appealing for non-invasive continuous monitoring [3]. Measuring stress with an in-ear device opens the door to personalized interventions, such as adaptive sound attenuation for people with decrease sound tolerance [4]. Cardiac signals can be captured using different sensors placed inside the ear canal. Most research in this area focused on the use of in-ear photoplethysmography (PPG) [5]. An In-Ear Microphone (IEM) was used in [6], and more recently, [7] used audio-plethysmography with an ultrasonic microphone integrated within Active Noise Cancellation (ANC) earphones. In the existing literature, in-ear heartbeat sounds have primarily been used for heart rate calculation, with the exception of [7], which focused on HRV computation. Although the [7] showed accurate results for HRV calculation, the method requires an ultrasonic microphone, as well as a speaker that actively sends probing signals at ultrasonic frequencies. It is of interest to develop a technique that can achieve accurate HRV calculation, strictly relying on sensors already integrated into hearables, like IEMs, which serve multiple functions including speech capture [8].

One approach to identify and analyze heartbeats is to use a template representing the typical shape and features of a normal heartbeat. Template-based algorithms for detecting heartbeats and signal quality assessment have been applied across different sensor modalities, such as radar, PPG and electrocardiography (ECG) [9], [10]. However, to the best of the authors' knowledge, this approach has not been exploited for acoustic heartbeat signals captured with an IEM. This paper introduces a template-based algorithm specifically

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designed for IEM heartbeat signal detection and evaluates its performance against existing algorithms applied to IEM heartbeat data reported in the literature.

II. METHODOLOGY

A. Algorithm

1) *Signal Pre-processing*: The signal is filtered with a 2nd order bandpass Butterworth filter with a low-cut frequency set at 0.5 Hz and a high-cut frequency at 25 Hz. The filtered signal is then resampled from 1000 Hz to 100 Hz.

B. Template generation

A median template of the heartbeat is generated by selecting segments around high-confidence peaks. These segments are determined based on relative peak heights and relative IBIs, with the goal to identify typical heartbeats within the signal to generate the template.

1) *Computing maximum and minimum heights*: First, a window of analysis is established based on a minimum heart rate, within which the algorithm identifies maximum signal amplitudes as potential heartbeats. This window of analysis is set to 1500 milliseconds, corresponding to an assumed minimum heart rate of 40 Beats Per Minute (BPM). The maximum in each window is taken as the peak. To eliminate outliers, these potential peaks are first normalized with Median Absolute Deviation (MAD) normalization (i.e., subtracting the median and dividing by the MAD).

Peaks with normalized values exceeding 2 or falling below -2 are then discarded. If the number of remaining peaks is fewer than 5, the peaks are prioritized based on their absolute standardized values, focusing on those that closely resemble the most common peak height. Subsequently, the minimum and maximum height of these selected peaks is computed.

2) *Peak detection and filtering*: Peaks are detected with a peak detection function using SciPy version 1.9.1 [11]. Peaks are removed if they are not within the minimum and maximum height boundaries calculated in the previous step. A second filtering step is applied based on the intervals between peaks: initial intervals are kept if they fall within 300 and 1500 milliseconds (based on an assumed minimum heart rate of 40 BPM and maximum heart rate of 200 BPM). Then, these initial intervals are standardized by subtracting the median and dividing by the median absolute deviation. A 3-point moving average is taken of the absolute values of the standardized intervals, and intervals are only kept if the averaged standardized value is less than 2.5.

3) *Segmentation*: Finally, the timestamps corresponding to the remaining intervals are selected to generate the template (in this work, the timestamps of the second peak of each interval were chosen). For each timestamp, 300 ms before the timestamp and 300 ms after the timestamp is selected from the filtered resampled signal. The median waveform is then computed with these segments to obtain the template.

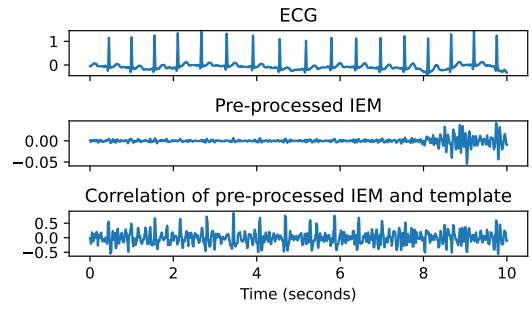


Fig. 1. Correlation signal derived from pre-processed IEM. The corresponding ECG signal is also shown as a reference of where peaks should be visible.

C. Correlation and Peak Detection

The generated template is then correlated with the pre-processed signal to identify potential heartbeat occurrences.

1) *Correlation*: Time-windows of 600-millisecond duration are taken for calculating the normalized correlation with the template. This is repeated for each subsequent sample within the pre-processed signal. The resulting correlation values, along with their corresponding time points, are compiled into arrays (as shown in Fig. 1). High correlation values should indicate a strong similarity to the median template, suggesting the presence of a heartbeat at those times.

2) *Peak detection from correlation*: Neurokit2's default ECG processing method is used with the correlation signal [12]. After high-pass filtering the signal with a cutoff of 0.5 Hz, the signal's gradient is computed and smoothed, in order to identify major changes indicative of potential heartbeats. The smoothed gradient is thresholded dynamically to distinguish large slopes, minimizing the impact of small fluctuations. Then, potential heartbeat segments are identified when the smoothed gradient exceeds this threshold. For each valid segment, the algorithm searches for the peak using prominence-based detection, ensuring a minimum interval of 300 ms between peaks to avoid false detection of multiple peaks in a short span. The artifact-correction algorithm described in [13] is then applied to the detected peaks.

D. Anomaly Removal and Peak Fixing

Detected peaks undergo additional processes to remove anomalies based on their correlation with the template and IBIs.

1) *Anomaly Removal*: Peak times are first removed based on the amplitude of peaks within the correlation signal. Only peaks surpassing a predetermined amplitude threshold are considered for further analysis. If the count of these peaks is less than a minimum required number of retained peaks (set to 20), the peaks with the highest amplitudes are selected to satisfy this criterion. Then, an interval-filtering procedure identical to the second filtering step in Section II-B2 is applied to the IBIs calculated from the remaining peaks. In cases where anomaly removal results in fewer peaks than the

minimum number of retained peaks required, the intervals that are least anomalous are taken until the minimum number of peaks is achieved. The intervals are subsequently converted back to peak times.

2) *Peak Fixing*: The algorithm then fills gaps and fixes detected peaks using a modified version of the peak-fixing function available in the Neurokit library [12]. To address gaps between peaks surpassing the maximum IBI, an estimation of missed peaks is conducted. This involves calculating a localized mean interval based on the nearest specified number of intervals, typically set at eight. Subsequently, the number of missing values required to fill the large gap is determined by dividing the large interval by the localized mean IBI and rounding up. Finally, these missing values are interpolated using Akima interpolation to bridge the gap and restore the missing peaks.

E. Performance evaluation

1) *Dataset*: The dataset used in this experiment was collected with the intent of recording acoustic and motion data using a sealing in-ear device featuring IEMs for heartbeat monitoring in an occluded ear canal along with concurrent ground truth ECG measurements. This dataset is recorded with various earpiece prototypes, all containing an IEM, Outer-Ear Microphone (OEM) and an In-Ear Receiver (IER). The data collection setup was carried out in an audiometric booth containing powered loudspeakers. Participants were instructed to wear the earpiece. Ground truth ECG signals were captured using the BioHarness 3.0 (Zephyr, Boulder, Colorado, USA), a lightweight, chest-worn strap that incorporates ECG and breathing rate sensors, which is strapped onto the user's chest.

To ensure a proper passive acoustic attenuation of the earpiece with the ear canal, three earpiece "fit-checks" were performed. This done by playing white noise at approximately 93 dB(SPL) on the loudspeakers and calculating the difference in sound pressure levels between the OEM and the IEM. The attenuation across octave-band frequencies was calculated and if the participant had an attenuation lower than 12 dB in the 125 Hz band, the participant was asked to improve their fit by removing and/or re-positioning the earpiece. Audio was recorded using the IEM and OEM at 48 000 Hz and then downsampled to 1 000 Hz for the purpose of this study.

Recordings from 39 participants (20 male, 19 female) in 2 different noise conditions, quiet and noisy, were included in this study. In the quiet condition, participants were either asked to breathe normally or hold their breath. In the noisy condition, participants were asked to breathe normally while noise was played directly in the IER or on the loudspeakers of the room.

2) *Evaluation Metrics*: The Mean Absolute Error (MAE) was calculated between the detected IBIs and ground truth IBIs. The IBIs were calculated with the time difference between consecutive detected heartbeat timestamps and consecutive ground truth heartbeat timestamps from the ECG signal. Both IBI signals were then interpolated at 2 Hz to ensure they

TABLE I
MEAN ABSOLUTE ERROR OF INTERBEAT INTERVALS (MEAN $\pm$ SD)

Length	Martin et al. [6]	Proposed without Template	Proposed with Template
15	15.11 $\pm$ 11.83	14.89 $\pm$ 15.26	9.22 $\pm$ 10.21
30	10.49 $\pm$ 8.68	7.38 $\pm$ 9.08	5.64 $\pm$ 7.13
60	10.53 $\pm$ 8.24	7.21 $\pm$ 7.17	5.57 $\pm$ 6.91
120	10.53 $\pm$ 8.91	7.03 $\pm$ 6.79	5.29 $\pm$ 6.9
180	10.35 $\pm$ 7.74	6.85 $\pm$ 6.67	4.92 $\pm$ 5.54
240	10.94 $\pm$ 8.17	6.89 $\pm$ 6.45	4.99 $\pm$ 5.15
300	10.43 $\pm$ 8.02	6.63 $\pm$ 6.3	4.68 $\pm$ 5.24

could be directly compared on a consistent time scale, and the mean of their differences was calculated.

3) *Analysis*: To better understand the effect of segment length on the performance of the template based algorithm, sections ranging from 15 seconds to 5 minutes, were selected.

The algorithm's performance was analyzed using three different methods of heartbeat extraction. The detected peak times were converted into IBIs and were compared against the ground truth IBIs derived from the ECG signal. For benchmarking purposes, a basic envelope-detection algorithm for heartbeat extraction that pioneered this field of in-ear biosignal monitoring, was re-implemented following the methods described in [6]. Additionally, to better understand the impact of taking the correlation of the template, the peak detection (described in Section II-C2), followed by the anomaly removal and peak fixing steps (described in Section II-D), was directly applied to the pre-processed signal (described in Section II-A1) instead of the correlation of the template. This is referred to in the text as "Proposed without Template". These two benchmark algorithms were compared to the algorithm following all the steps explained in this paper ("Proposed with Template").

III. RESULTS AND DISCUSSION

Table I presents the Mean Absolute Error (MAE) for different algorithms as well as different segment lengths. Comparing the proposed algorithm with the benchmark algorithm from [6], we see a substantial decrease in the MAE that is amplified by the increase in segment length. Moreover, the addition of the median template further improves the performance. It is likely that the increase in segment length improves various steps of the proposed algorithm, such as the anomaly removal and peak fixing, as well as the estimation of the median template, as shown in Fig. 2. Fig. 3 illustrates the performance of our proposed algorithm compared to the other algorithms, using two signals: a good example and a bad example with low Signal-to-Noise Ratio (SNR).

Given that the recommended duration for short-term HRV analysis [14] is 5 minutes, and that performance improves with larger windows, it would be advantageous to use the entire 5 min window for extraction of IBIs. However, for real-time extraction of "ultra short-term" HRV features using shorter windows, future work could focus on optimizing the algorithm to retain parameters following an initialization phase. Our

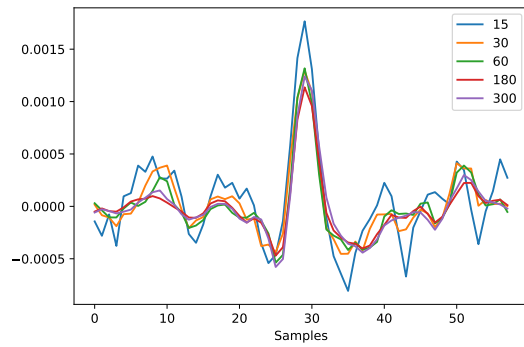


Fig. 2. Comparison of template across multiple signal lengths (in seconds)

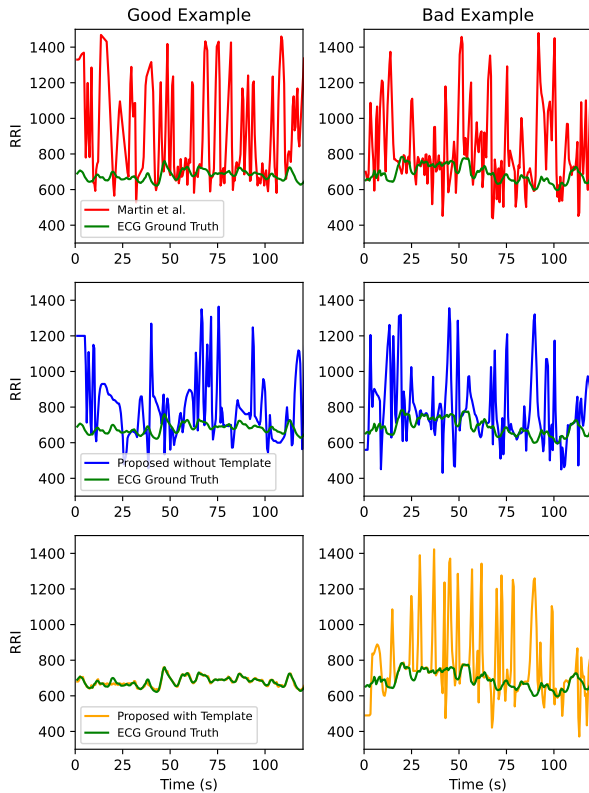


Fig. 3. Comparison of IBIs using different heartbeat extraction algorithms for a good and a bad example. The input segment was 5 minutes long; however, the first 2 minutes are shown here.

evaluation focused on the interpolated IBIs. For some HRV features, such as the root mean square of successive differences, it may be advantageous to simply discard segments with lower confidence instead of discarding them and filling the resulting gaps [15]. Future work could explore whether skipping the peak fixing step and using only the IBIs with higher confidence could improve the accuracy of certain HRV features.

IV. CONCLUSIONS

The results of this study support the effectiveness of template-based algorithms for extracting IBIs from in-ear

audio data. This approach paves the way for personalizing existing hearable features, such as noise cancellation, particularly for individuals with decreased sound tolerance. By tailoring these technologies to individual needs, we can enhance user comfort and overall listening experience.

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